

Tim N. Ortkamp

Doctoral Researcher in Applied Mathematics & Medical Physics

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EDUCATION & RESEARCH

Visiting Researcher	2024 & 2026
Department of Radiation Oncology, Massachusetts General Hospital Boston, USA	

Doctoral Researcher	2021–2026
Scientific Computing Center, Karlsruhe Institute of Technology & Department of Medical Physics in Radiation Oncology, German Cancer Research Center Karlsruhe / Heidelberg	

Master of Science (M.Sc.) – Industrial Engineering	2018–2021
Karlsruhe Institute of Technology Karlsruhe	

Bachelor of Science (B.Sc.) – Industrial Engineering	2014–2018
Karlsruhe Institute of Technology Karlsruhe	

Abitur (University Entrance Qualification)	2006–2014
Viktoriaschule Darmstadt Darmstadt	

PROFESSIONAL EXPERIENCE

Student Assistant	2016–2021
Institute for Operations Research, Karlsruhe Institute of Technology Karlsruhe	

Intern in Data Science	2019
Smart Data Analytics / MES & IIoT, Dürr Systems AG Bietigheim-Bissingen	

Intern in Supply Chain Consulting	2015 & 2017
Strategy and Concept Planning, Miebach Consulting GmbH Frankfurt am Main	

PUBLICATIONS, PREPRINTS & OTHER WORKS

- [1] **T. Ortkamp**, H. Sallem, S. Harrabi, M. Frank, O. Jäkel, J. Bauer, N. Wahl. Direct optimization of the probability of lesion origin in proton treatment planning for low-grade glioma patients. *Medical Physics*, 2026. DOI: [10.1002/mp.70395](https://doi.org/10.1002/mp.70395)
- [2] **T. Ortkamp**, H. Sallem, S. Harrabi, M. Frank, O. Jäkel, J. Bauer, N. Wahl. Direct optimization of the probability of lesion origin in proton treatment planning for low-grade glioma patients. *arXiv preprint*, 2025. [arXiv:2506.13539](https://arxiv.org/abs/2506.13539) [physics.med-ph]
- [3] L. Voss, A. Neishabouri, **T. Ortkamp**, A. Mairani, N. Wahl. BayesDose: Comprehensive proton dose prediction with model uncertainty using Bayesian LST. *arXiv preprint*, 2023. [arXiv:2307.01151](https://arxiv.org/abs/2307.01151) [physics.med-ph]

Lecture Notes:

T. Ortkamp, M. Coblenz, O. Grothe. *Analyse multivariater Daten* (Analysis of Multivariate Data), 2021.

CONFERENCES & WORKSHOPS

- ESTRO Physics Workshop (Toulouse, 2025)
- ICCR 20 (Lyon, 2024, oral presentation & conference paper: [HAL: hal-04720234](#) (p. 841–844))
- EUROPT 21 (Lund, 2024, oral presentation)
- PTCOG 62 (Singapore, 2024, oral presentation)
- EUCCO 6 (Heidelberg, 2023, oral presentation)
- AAPM 65 (Houston, 2023, oral presentation)
- PTCOG 61 (Madrid, 2023, oral presentation)

OPEN-SOURCE SOFTWARE CONTRIBUTIONS

[pyanno4rt](#) (Lead Developer)

Python library for conventional and outcome prediction model-based inverse photon and proton treatment plan optimization.

[seamaze](#) (Lead Developer)

Python library for classical, limited-memory, and dynamical low-rank CMA-ES.

[pyRadPlan](#) (Contributor)

Radiotherapy treatment planning toolkit designed for interoperability with matRad.

TEACHING & ACADEMIC SUPERVISION

Co-supervisor

[10/2022–03/2023](#)

Seminar “Optimierung in der KI” (Optimization in AI), Scientific Computing Center, Karlsruhe Institute of Technology

Karlsruhe

Co-instructor & Assistant lecturer

[04/2022–08/2022](#)

Course “Optimierungstheorie” (Optimization Theory), Scientific Computing Center, Karlsruhe Institute of Technology

Karlsruhe

Instructor

[10/2019–03/2020](#)

Course “Analyse multivariater Daten” (Analysis of Multivariate Data), Institute for Operations Research, Karlsruhe Institute of Technology

Karlsruhe

Tutor

[10/2016–02/2017](#)

Course “Statistik II” (Statistics II), Institute for Operations Research, Karlsruhe Institute of Technology

Karlsruhe

Student Supervision

- *Proximale Algorithmen zur Optimierung von nicht-differenzierbaren Problemen in der Radiotherapie* (Proximal Algorithms for the Optimization of Non-Differentiable Problems in Radiotherapy) (Bachelor’s thesis, V. Deisner, 2025)
- *Machine Learning Model-Based Radiotherapy Treatment Plan Optimization Using Limited Memory Matrix Adaption Evolution Strategies* (Bachelor’s thesis, M. Müller, 2024)

- *Inverse Radiotherapy Treatment Planning using Machine Learning Outcome Prediction Models* (Student assistance, M. Müller, 2022–2024)
- *Machine Learning for Radiotherapy Outcome Prediction based on Imbalanced Data* (Bachelor's thesis, T. Thost, 2023)
- *Restringierte Optimierung von maschinellen Lernmodellen zur Vorhersage des Behandlungserfolgs in der Radiotherapie* (Constrained Optimization of Machine Learning Models for Outcome Prediction in Radiotherapy) (Bachelor's thesis, K. von Gaisberg, 2023)
- *Multikriterielle Optimierung und ihre Anwendung in der inversen Radiotherapieplanung* (Multi-Objective Optimization and its Application in Inverse Radiotherapy Treatment Planning) (Bachelor's thesis, L. Gseller, 2023)
- 2 co-supervisions at the German Cancer Research Center

SKILLS, AWARDS & INTERESTS

Languages

- *German*: Native
- *English*: Fluent
- *Swedish*: Basic

Awards

- Certificate for good teaching
Course “Analyse multivariater Daten” (Analysis of Multivariate Data), Karlsruhe Institute of Technology, 2020

Software & Programming

- *Languages*: Python, MATLAB, R
- *Tools & Markup*: Git, L^AT_EX, HTML/CSS

Personal Interests

- *Sports*: Cycling, Running, Soccer, Hiking
- *Music*: Piano, Synthesizer, Guitar, Songwriting, Audio recording/production